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LTE NETWORK – AN EVOLUTION IN SCADA COMMUNICATION

Presented By

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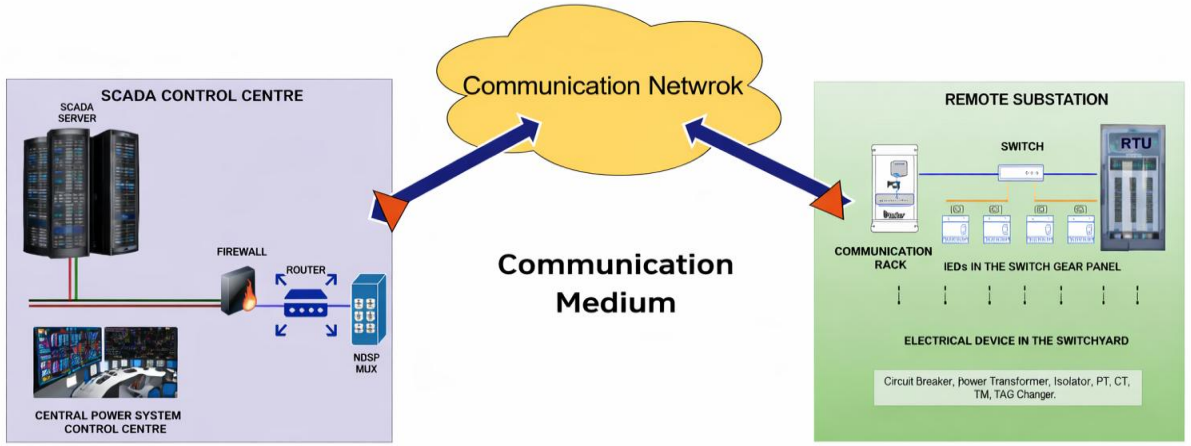
INTRODUCTION

A **Supervisory Control and Data Acquisition (SCADA)** system relies heavily on a robust **communication system** to ensure seamless data exchange between remote field devices, controllers, and central control units.

Advantages:

- ❑ Improve efficiency, reduce downtime, and enhance overall productivity
- ❑ Ensure high reliability, availability and security
- ❑ Overall improve Operational Excellency

Real-time SCADA Communication



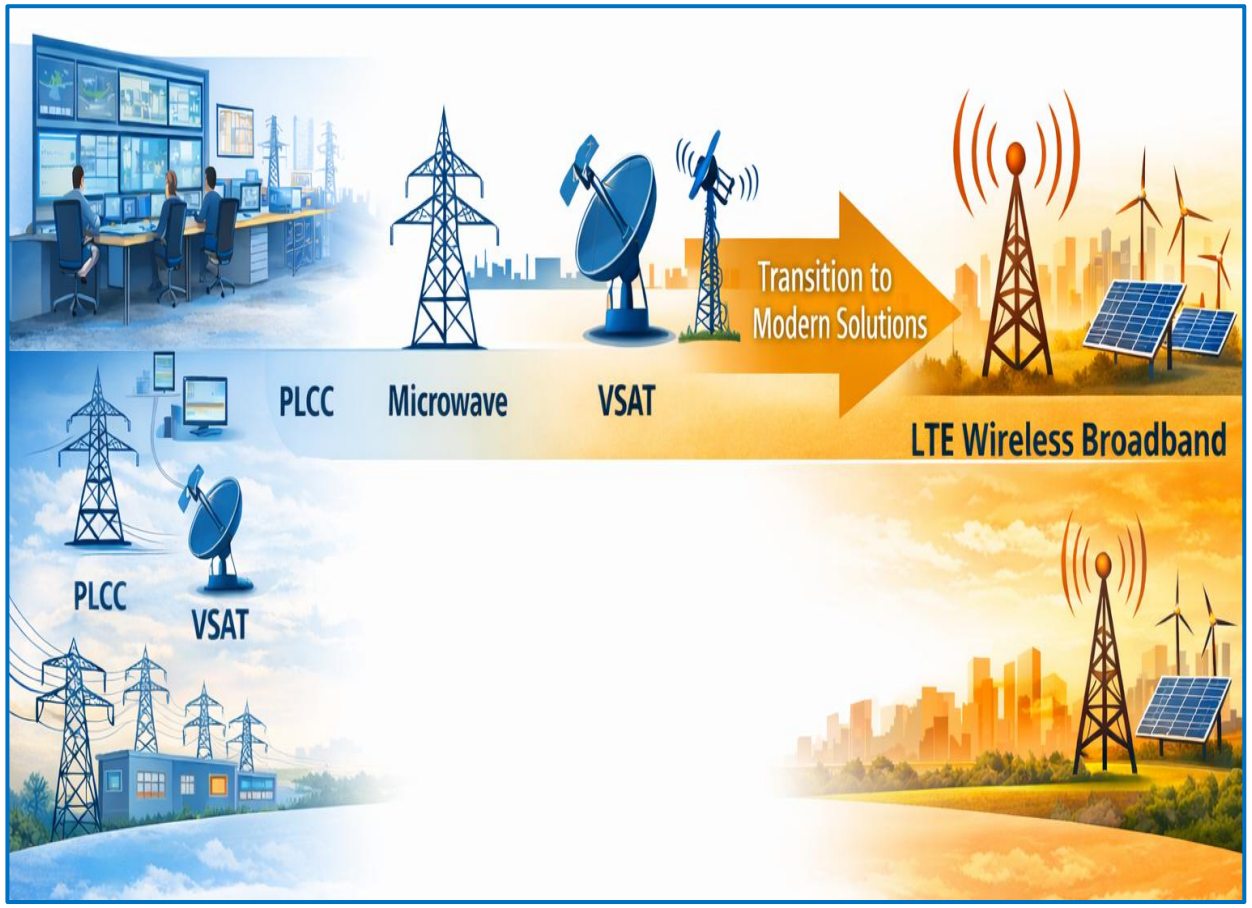
The limitations of traditional SCADA communication methods. (leased lines, OFC, RF, and satellite)

- The limitations of traditional SCADA communication methods
- Limited bandwidth
- Higher latency
- Expensive infrastructure & maintenance
- Limited scalability
- Lower flexibility for modern smart grid applications

CONTEXT

CONTEXT :

- Traditional SCADA communication relied on PLCC, RF, Microwave, and Leased Lines
- Designed for low-bandwidth telemetry and basic control
- Limited scalability, higher latency, and infrastructure dependency
- Emergence of Smart Grids & Digital Substations increased data requirements
- Integration of DER, IoT devices, Smart Meters, and Automation raised communication demand
- Need for secure, scalable, high-speed, low-latency IP-based networks
- LTE technology emerges as a reliable broadband solution for modern SCADA systems



OBJECTIVE :

- To transform SCADA communication into a secure, scalable, and high-performance, LTE-based infrastructure for modern smart grid operations.

SOLUTION BRIEF

What is LTE?

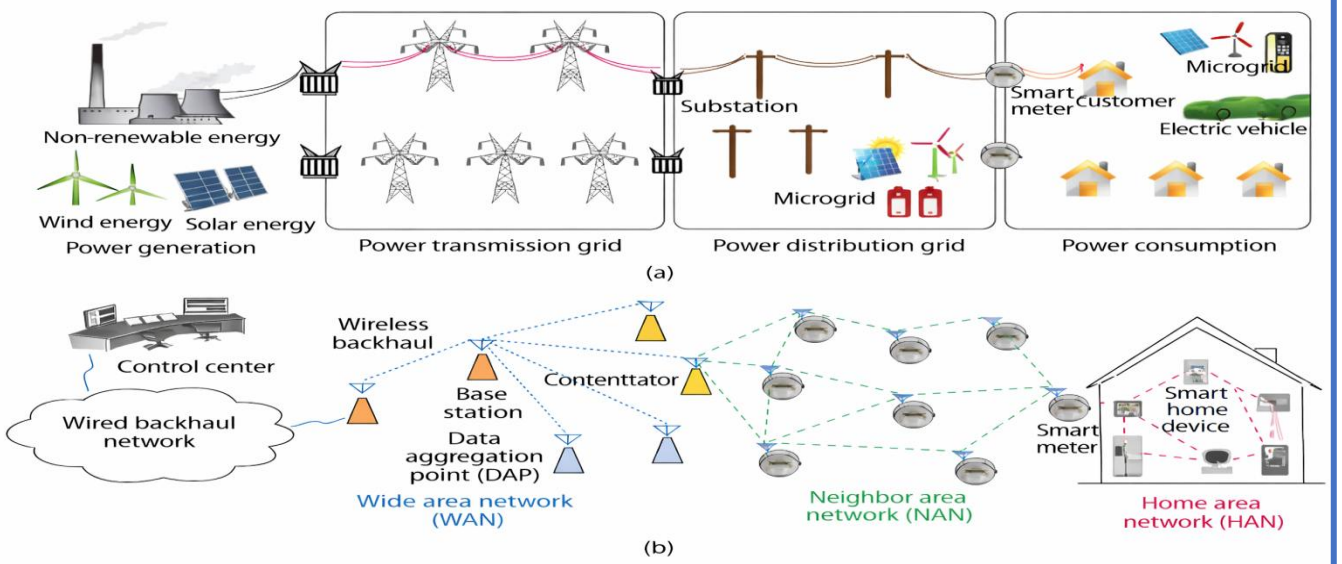
A 4G IP-based broadband wireless technology delivering 100–300 Mbps, low latency, and high reliability for real-time data communication.

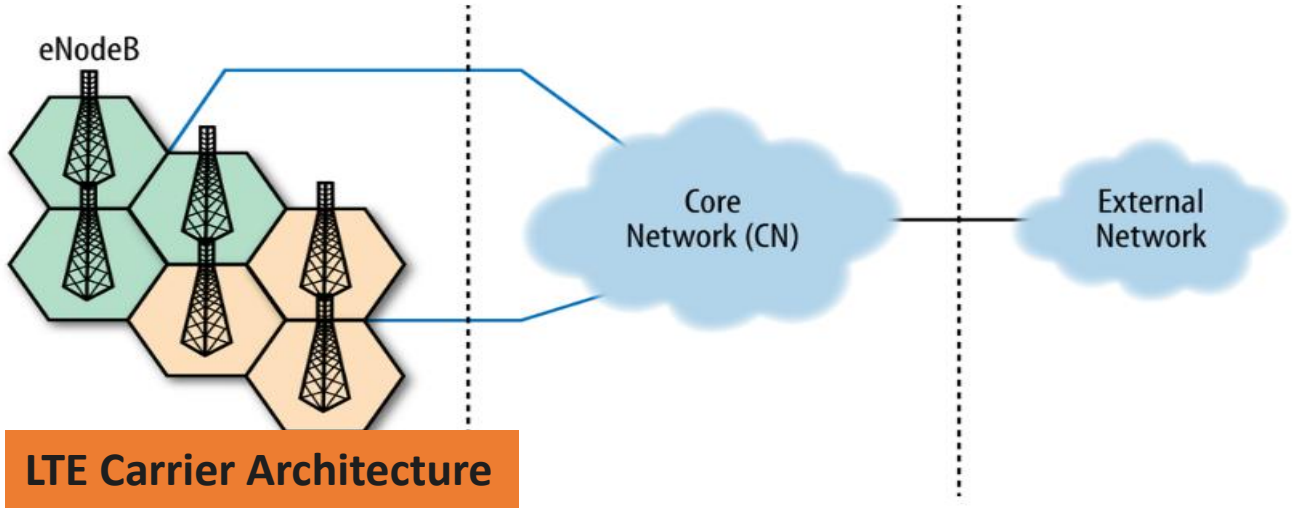
Key Technology Features:

- Fully IP-based packet network
- OFDMA & MIMO for high data throughput
- Low latency (10–20 ms) for real-time control
- QoS prioritization for critical SCADA traffic
- SIM-based authentication & IPSec security

Advantages for SCADA Systems

- High bandwidth vs. PLCC/RF/Microwave
- Faster fault detection & restoration
- Scalable for Smart Grid & Digital Substations
- Cost-effective
- Secure, reliable & future-ready communication platform





LTE Carrier Architecture

METHODOLOGY

METHODOLOGY IN SCADA COMMUNICATION OVER LTE NETWORK :

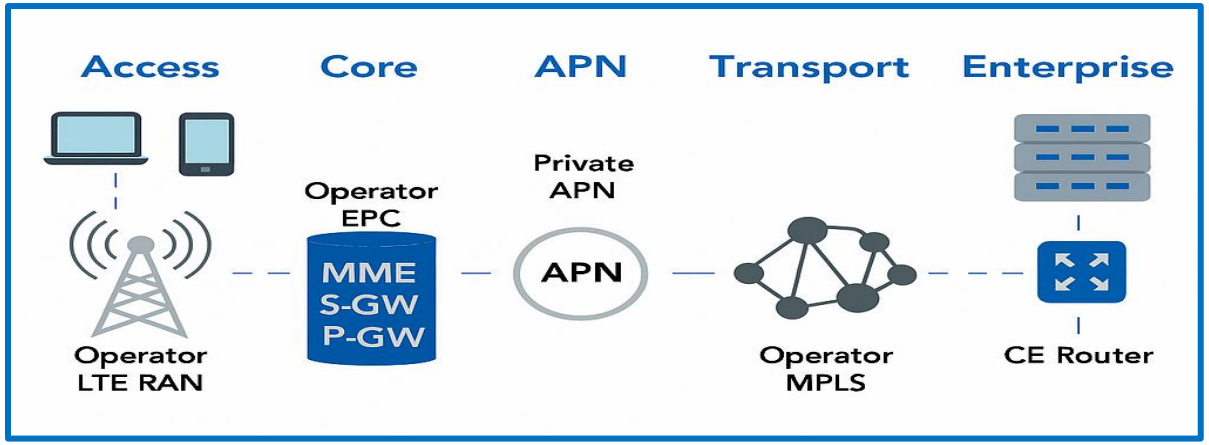
SCADA Communication over LTE can be deployed using following methods:

- **Public LTE Network:** Utilizes the existing cellular network provided by telecom operators.
- **Private LTE Network:** A dedicated LTE infrastructure for better security and performance.
- **Hybrid LTE Network:** Combines private LTE in critical areas with public LTE for remote locations.

Feature	Private LTE	Public LTE
Ownership	Enterprise	Telecom Operator
Data Security	Very high (on-premise)	High but external
Coverage	Limited to campus	Nationwide
Reliability	Very high	Varies
Congestion	No	Yes (shared users)
Customization	Full	Limited
Mobility	Excellent	Excellent
Cost	Capex + Minor OPEX	Subscription OPEX

OBJECTIVE :

- The deployment model is selected based on **security, reliability, coverage, and operational requirements** to ensure secure and scalable SCADA communication.



IMPLEMENTATION PHILOSHOPHY

IMPLEMENTATION PLAN FOR SCADA COMMUNICATION OVER LTE NETWORK:

The following three types of scenarios can be deployed to ensure secure and reliable communication:

Scenario #1:
Existing RF Link / OFC Link as Primary and LTE Network (SIM) as Secondary

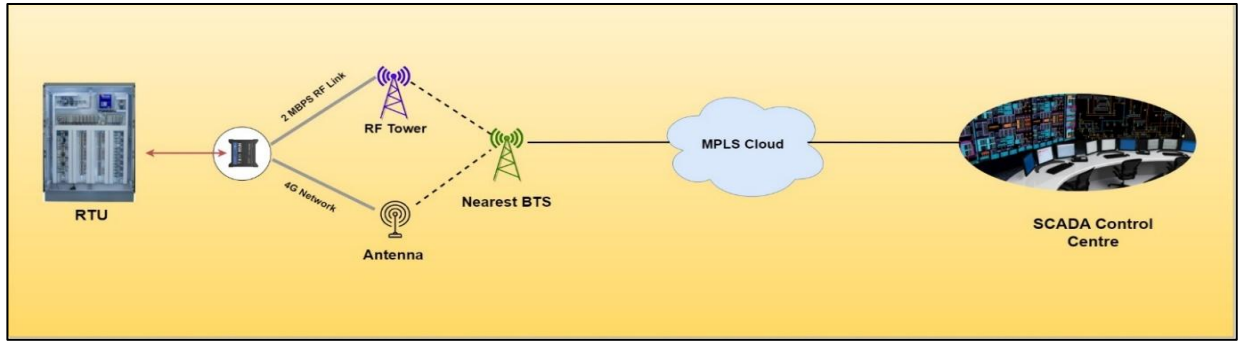
Scenario #2:
Dual SIM-Based Communication for Redundancy.

Scenario #3:
Hybrid Model with RF / Fiber Link as Primary and Dual 4G SIM as Standby.

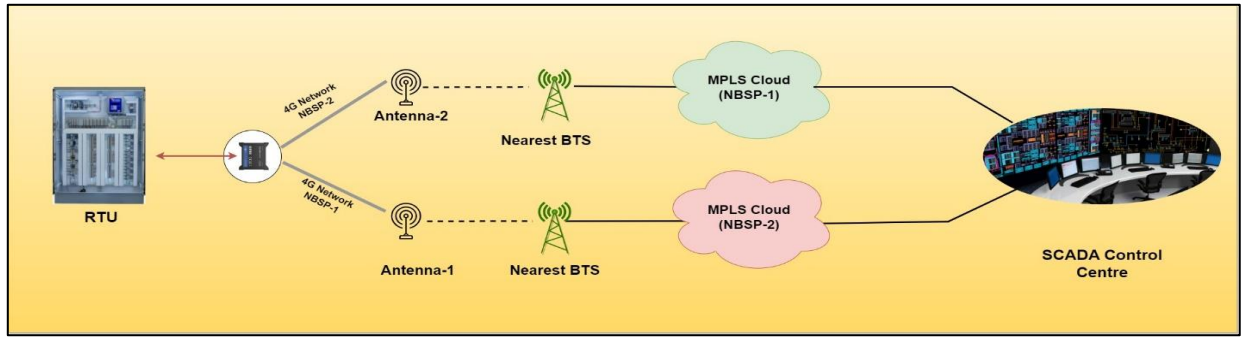
OBJECTIVE :

Enhance communication reliability, minimize downtime, and ensure continuous SCADA operation.

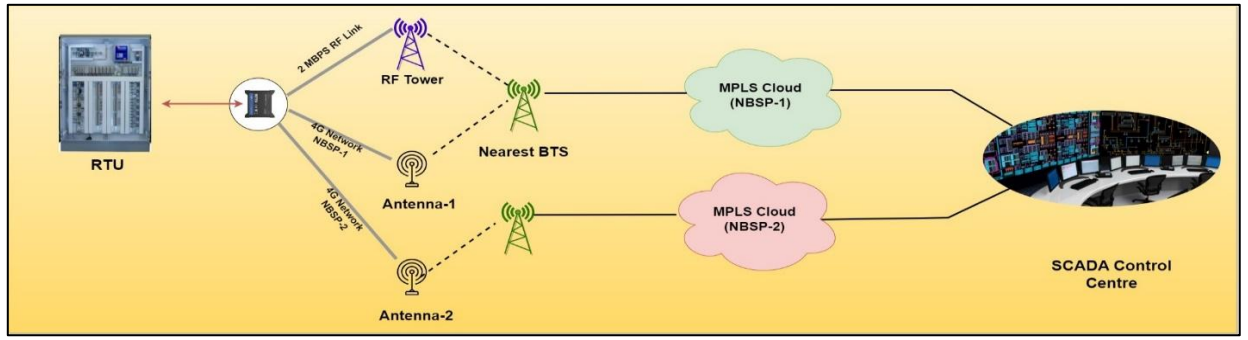
Scenario #1:



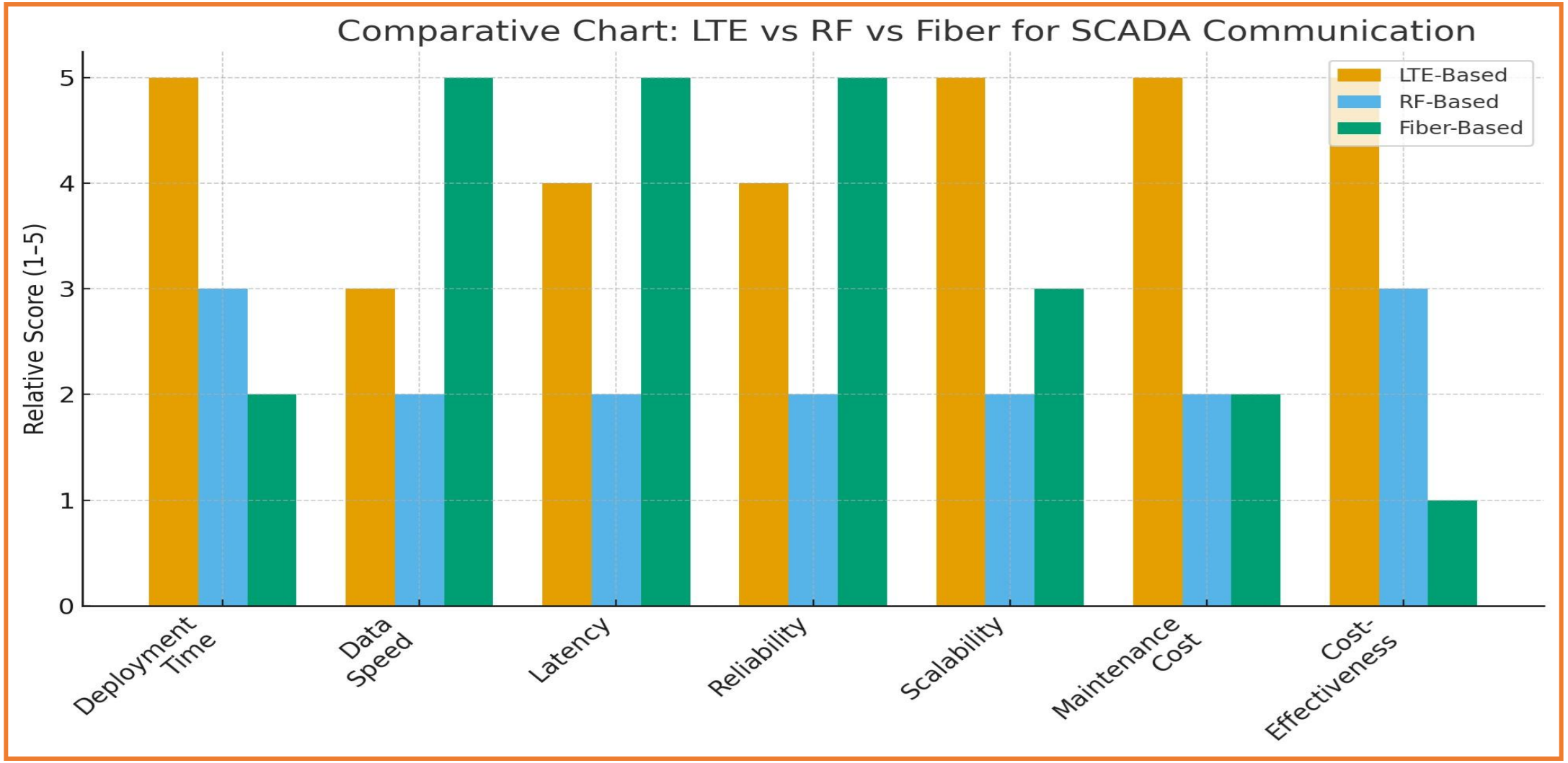
Scenario #2:



Scenario #3:



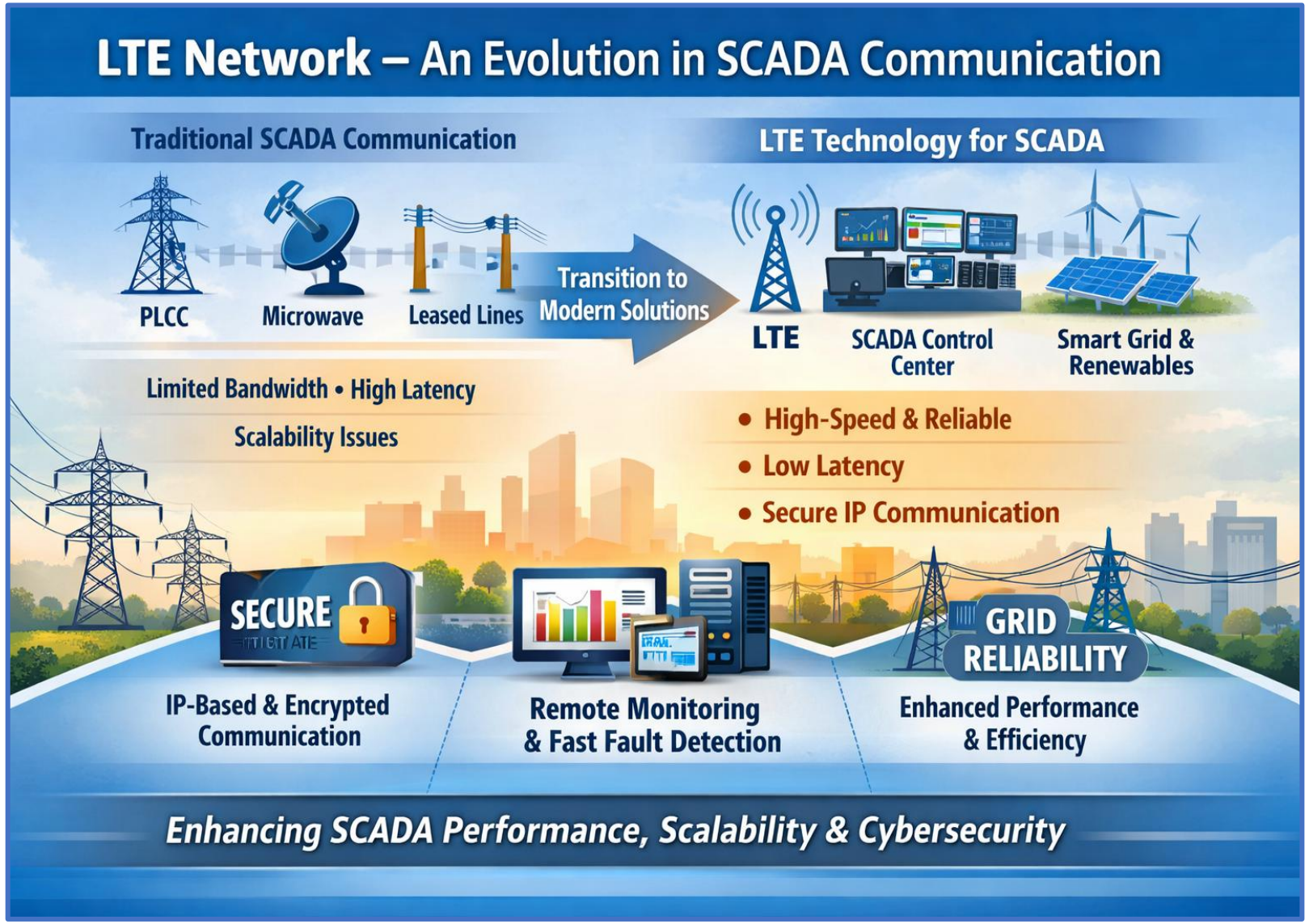
CONCEPTUAL ANALYSIS



CONCLUSION

LTE-based SCADA communication represents a modern, efficient, and future-ready solution for power system networks. Advantages as follows:

- Cost-effective and scalable solution for modern power networks.
- Reducing CAPEX and OPEX.
- Ensures high availability, low latency, and improved reliability.
- Advanced security features support secure OT communication.
- A strong alternative to traditional PLCC, microwave, and leased-line systems.



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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)